Question Bank 2025 June with Anwser

Question 1

Which statement is true?

a. The Notes section of financial reports usually contains the numerical value of the value drivers.

b. Before financial analysis, the historical accounting statements should be modified for three reasons: (1) Suspected accounting mistakes, (2) Recent changes in the accounting regulations, (3) Changes in macro conditions.

c. We do not simply average the different results from multiple valuation approaches to arrive at a final firm value.

d. Synergies from cost-cutting in M&A transactions are harder to capture than those linked to growth.

Explanation

Different valuation methods rely on different assumptions and have different strengths and weaknesses. Analysts therefore use judgement instead of mechanically averaging all valuation results.

Question 2

What category of financial ratios is based exclusively on balance sheet data?

a. Liquidity ratios

b. Classic leverage ratios (like in Miller-Modigliani theories)

c. Interest coverage ratios

d. Working capital turnover ratios

Explanation **not sure which one is correct**

Liquidity ratios (e.g. Current Ratio, Quick Ratio) use only balance-sheet items. Interest coverage and turnover ratios require income statement information.

Question 3

Which of the following qualifies as a standard contractual covenant?

a. No dividend payment is allowed

b. Minimum of 1.3 DSCR

c. Material adverse change

d. Ownership clause

Explanation

A contractual covenant is a clause written directly into the loan agreement that governs ownership, control, or borrower behavior.

An ownership clause specifies restrictions related to changes in ownership structure and therefore qualifies as a standard contractual covenant.

A DSCR requirement is instead a financial covenant because it imposes a numerical financial ratio requirement.

Question 4

Which statement below is correct?

a. Most individual investors have portfolios containing more than 10 stocks.

b. The CAPM assumption that all investors have homogeneous expectations accurately reflects real-world conditions.

c. In practice, pinpointing efficient portfolios is very challenging since accurately measuring a portfolio's expected return and return standard deviation is difficult.

d. Investors exhibiting familiarity bias are likely to invest in exotic shares.

Explanation

Efficient portfolios depend on expected returns and risk estimates, both of which are difficult to estimate accurately in reality.

Question 5

Read all the possible answers below!

A leasing has to be treated as a finance lease if...

a. The lease term is 75% or more of the estimated economic life of the asset.

b. The title to the property transfers to the lessee at the end of the lease term.

c. All the other three answers are correct.

d. The lease contains an option to purchase the asset at a bargain price that is substantially less than its fair market value.

Explanation

All three listed criteria are classic indicators that substantially all risks and rewards of ownership have been transferred to the lessee.

Question 6

Suppose a firm operates like a perpetuity.

EBIT = 400, $r_A = 12.2\%$, $r_D = 5\%$, effective tax rate = 10%, RONIC = 12%, $g = 4\%$, $D/V = 40\%$.

You have to value a 40% stake in equity.

A 10% discount for high dependence on a single key person is justified.

As one dominant investor owns the remaining 60%, you decide to apply a 10% minority discount.

Which statement is true?

a. Value of the 40% stake is almost 720.

b. Value of the 60% stake is almost 1040.

c. Value of the equity is 3000.

d. Value of the firm is 4500.

Explanation **calculation will be added later**

Question 7

Which of the following statements is correct?

a. When assuming ROIC equals WACC during the terminal value period, we expect the terminal value to be equal to invested capital (IC).

b. A firm's required reinvestment ratio depends on its future growth targets and the expected average return on its investments.

c. We estimate the target firm's required equity return by averaging the unleveraged cost of equity of its peer companies.

d. It is generally assumed that the unleveraged cost of capital is equal to the WACC during the terminal value period.

Explanation
Economic profit is defined as

$$Economic\ Profit = (ROIC - WACC) \times IC$$

If

$$ROIC = WACC$$

then

$$Economic\ Profit = 0$$

and future investments generate zero net present value.
Consequently:

$$Terminal\ Value = Invested\ Capital$$

This is a standard result in DCF valuation and economic-profit valuation frameworks.

Question 8

What are the typical features of project loan financing?

a. Long maturities

b. Contract based financing

c. Financing is on a non-recourse basis

d. The other three answers are all correct.

Explanation
Project finance is typically:

- Long-term
- Based on project contracts
- Non-recourse or limited-recourse

Hence all three statements are true.

Question 9

Which statement is true?

a. One of the key ideas of Prospect Theory is that people do not know probabilities exactly and hence overestimate small probabilities. This can be modeled by a probability weighting function.

b. Allais' Paradox shows an inconsistency of actual observed choices with the predictions of Expected Utility Theory.

c. Allais' Paradox tells us that Expected Utility Theory cannot explain all observed decisions.

d. If the market is incomplete, then asset prices are unique.

Explanation **not sure b or c**
Allais' Paradox is a famous empirical violation of Expected Utility Theory and motivated the development of alternative behavioral theories.

Question 10

Which of the following statements is incorrect?

a. Ambiguity aversion means that people do not like unknown probabilities.

b. Because markets are complete, Bloomberg offers about four different models to price complex options, each producing a different price.

c. Methods exist to quantify a decision maker's utility function.

d. The mean-variance utility function only considers variance and fails to account for higher moments, thereby inadequately addressing the risk of extreme losses (fat tails or black swan events).

Explanation
Market completeness implies a unique arbitrage-free price. The existence of multiple pricing models is not caused by market completeness. Therefore this statement is incorrect.

Question 11

Which statement below is incorrect?

a. The efficient frontier consists of all portfolios within the mean-variance opportunity set for which no other portfolio exists that offers both a higher expected return and lower risk.

b. When investors have mean-variance preferences with heterogeneous beliefs, better informed investors can detect and exploit alpha opportunities, but only at the expense of worse informed investors.

c. Trading driven by heterogeneous beliefs in financial markets can be summarized as: "I believe stocks will rise, you believe they will fall, so let's trade!"

d. If the equilibrium prices of all assets are unique, then the market is complete.

Explanation
A complete market implies unique arbitrage-free prices.
However, unique prices do not necessarily imply market completeness.
Therefore the implication is not reversible.

Question 12

Which of the following statements is incorrect?

a. Risk sharing on financial markets can be interpreted as follows: if I need the money later, you need it now, let's trade!

b. When the market is complete, asset prices are uniquely determined.

c. Prospect Theory can explain why individuals exhibit risk aversion for gains that have a moderate likelihood of occurring.

d. Allais' Paradox demonstrates that Expected Utility Theory fails to account for certain observed decision-making behaviors.

Explanation
The statement describes intertemporal exchange rather than risk sharing.
Risk sharing is better summarized as:
"I dislike a particular risk and you are willing to bear it, so let's trade."

Question 13

What is the correct comparison between the value of European, American, and Bermudan options, if all other parameters are the same?

a. European > American > Bermudan

b. American > Bermudan > European

c. European = Bermudan = American

d. Bermudan > European > American

Explanation
More exercise opportunities can never reduce option value.
Therefore:

$$V_{American} > V_{Bermudan} > V_{European}$$

Question 14

What does FRA stand for?

a. Forward Rate Agreement

b. Forward Rate Arbitrage

c. Futures Regression Analysis

d. Farrow-Ross-Abel model

Explanation
An FRA is an over-the-counter contract that locks in an interest rate for a future borrowing or lending period.

Question 15

Which of the following statements correctly describes the valuation of an interest rate swap?

a. There are two methods: forward method and FRA pricing method.

b. There are two methods: bond pricing method and futures pricing method.

c. There are two methods: bond pricing method and FRA pricing method.

d. There are three methods: forward method, bond pricing method and FRA pricing method.

Explanation
The standard approaches are:

- Bond pricing approach
- FRA decomposition approach

Both lead to the same theoretical swap value.

Question 16

Under what conditions is a margin call triggered, and what amount is the investor required to deposit in response?

a. If the balance in the margin account falls below the initial margin, he must pay up to the initial margin level.

b. If the margin account falls below the initial margin, he must pay up to the maintenance margin level.

c. If the balance in the investor's margin account declines below the maintenance margin requirement, the investor is obligated to deposit additional funds to restore the account balance to at least the maintenance margin level.

d. If the margin account balance falls below the maintenance margin threshold, the investor is required to replenish the account up to the initial margin level.

Explanation
A margin call occurs when account equity falls below the maintenance margin.
The investor must restore the balance to the initial margin level.

Question 17

Consider a portfolio consisting of two assets:

- one share of stock issued by company XXX,
- a zero-coupon bond that pays \$100 in one year.

Suppose this portfolio is currently trading with a bid price of \$171.22 and an ask price of \$174.11,
and the bond is trading with a bid price of \$95.5 and an ask price of \$97.9.
In this case, what is the no-arbitrage price range for XXX stock?

a. 70.22–73.11

b. 73.32–78.61

c. 75.82–76.11

d. It cannot be defined because its price depends on the riskiness of the stock, which is unknown.

Explanation
Using bid-ask arbitrage bounds:

$$S_{bid} = 171.22 - 97.9 = 73.32$$

$$S_{ask} = 174.11 - 95.5 = 78.61$$

Hence:

$$73.32 \leq S \leq 78.61$$

Question 18

Which type of derivative is challenging to price accurately using the Monte Carlo method?

a. European option

b. Barrier option

c. Asian option

d. American option

Explanation
American options allow early exercise.
Monte Carlo simulation naturally handles terminal payoffs but not optimal stopping decisions.
Therefore American options are the most difficult among the listed alternatives.

Question 19

The price of a down-and-out knock-out call option is 5.
The price of the vanilla call option with the same parameters is 8.
What is the fair price for the down-and-in knock-in call option with the same parameters?

a. 0

b. 3

c. 5

d. 13

Explanation
For barrier options:

$$\text{Vanilla} = \text{Knock-In} + \text{Knock-Out}$$

Therefore:

$$8 = 5 + \text{Knock-In}$$

$$\text{Knock-In} = 3$$

Question 20

In the Black-Scholes formula, what does $N(d_1)$ represent?

a. Lognormal probability density function evaluated at d_1

b. Standard normal probability density function evaluated at d_1

c. Lognormal cumulative distribution function evaluated at d_1

d. Standard normal cumulative distribution function evaluated at d_1

Explanation

$$N(d_1) = P(Z \leq d_1)$$

where

$$Z \sim N(0, 1)$$

Thus $N(d_1)$ is the cumulative distribution function (CDF) of the standard normal distribution evaluated at d_1.

Question 21

The Wiener process is Markovian. What is the meaning of this property?

a. Actually, the Wiener process is not Markovian.

b. The Wiener process does not have a drift.

c. On average, the Wiener process does not exhibit growth.

d. The Wiener process has no memory, meaning its future increments depend only on the current state, not on the past history.

Explanation
A Markov process satisfies:

$$P(X_{t+1}|X_t, X_{t-1}, ..., X_0) = P(X_{t+1}|X_t)$$

The future depends only on the current state and not on the entire path that led there. This property is often referred to as the "memoryless" property.

Question 22

What elements are included in the third pillar of Basel II regulation?

a. Supervisory review process

b. The stress testing methodology

c. Disclosure of the bank's risk assessment procedures

d. Regulatory capital calculation

Explanation
Basel II consists of three pillars:

- Pillar 1: Minimum Capital Requirements
- Pillar 2: Supervisory Review Process
- Pillar 3: Market Discipline through Disclosure

Therefore Pillar 3 focuses on transparency and disclosure requirements.

Question 23

A two-year project has the following possible outcomes:

- 90% probability of a \$5 million gain
- 9% probability of a \$10 million loss
- 1% probability of a \$20 million loss

Calculate the two-year Expected Shortfall (ES) at the 95% confidence level.

a. 12 million

b. 8 million

c. 20 million

d. 10 million

Explanation
The worst 5% outcomes consist of:

- 1% at -20 million
- 4% of the 9% observations at -10 million

Expected Shortfall:

$$ES_{95} = \frac{0.01 \times 20 + 0.04 \times 10}{0.05}$$

$$= \frac{0.6}{0.05} = 12$$

Thus:

$$ES_{95} = 12 \text{ million}$$

Question 24

If a financial institution applies the Foundation Internal Ratings-Based (F-IRB) approach for calculating capital requirements for credit risk, the parameter(s) it must estimate are:

a. EAD, M

b. LGD, EAD

c. LGD

d. PD

Explanation
Under the Foundation IRB approach:

- PD (Probability of Default) is estimated by the bank.
- LGD (Loss Given Default) is provided by regulators.
- EAD (Exposure at Default) is generally prescribed.

Therefore the bank estimates only PD.

Question 25

Which of the following does NOT qualify as an operational risk?

a. Conduct risk

b. IT breakdown

c. Increasing the central bank's base interest rate

d. Epidemic

Explanation
Operational risk arises from:

- Failed internal processes
- Human error
- System failures
- External events

Examples include cyberattacks, IT outages, fraud, conduct failures, and pandemics. Changes in central bank interest rates represent market risk rather than operational risk.